Jupiter Has Its Own, Busier Way of Blocking the Solar Wind

From: ScienceDaily Most Popular Mined: 2026-10-30 5 min read

NASA's Juno spacecraft has found that Jupiter, the solar system's largest planet, withstands the Sun's particle stream with a much more complex system of **plasma waves** and shock regions than Earth uses, in research led by scientists at the University of Iowa.

The Sun continuously releases a stream of electrically charged particles called the **solar wind**. When the **solar wind** approaches a planet with a magnetic field, it meets a boundary known as the **bow shock**.

At Earth, this boundary slows and heats incoming particles and redirects much of the **solar wind** around the planet. Without it, more harmful particles could reach the atmosphere. Jupiter also has a **bow shock**, but its enormous size, powerful magnetic field and gaseous makeup make it especially interesting to study.

Using measurements from Juno, researchers discovered that Jupiter relies on **plasma waves** with multiple frequencies to slow and heat incoming solar particles. These different frequencies, known as **harmonics**, create a more elaborate wave pattern than the one typically observed at Earth. It works a bit like music: several notes together sound richer than one note, and waves with multiple frequencies interact with particles in more complex ways, heating more of them and slowing them down. Earth's **bow shock**, by comparison, primarily uses waves concentrated around a single frequency.

Jupiter also has an extra line of defense. Before incoming particles reach the main **bow shock**, they can encounter smaller shock structures called **shocklets**, which help slow the **solar wind** first. Earth generally relies on its main **bow shock** alone. Earth does not need these extra steps because the impact is not as powerful, said Bill Kurth, an Iowa research scientist on the Juno mission since its 2011 launch.

Beyond Jupiter, the results could help scientists understand some of the most powerful events in the universe, including the extreme shock waves associated with exploding stars.

Word Treasure

solar wind -- A stream of electrically charged particles constantly released by the Sun.

bow shock -- The boundary where the solar wind first meets a planet's magnetic field and slows down.

plasma waves -- Waves travelling through plasma, a gas-like state of electrically charged particles.

harmonics -- Different frequencies of a wave that combine into a more elaborate pattern.

shocklets -- Smaller shock structures that slow the solar wind before it reaches the main bow shock.

Background you need

The Sun constantly sends out a stream of electrically charged particles, and this stream is called the solar wind.

Many planets have a magnetic field, an invisible force around the planet that can push back against charged particles from space.

When the solar wind runs into a planet's magnetic field, it forms a boundary that slows and heats the particles and steers most of them around the planet.

Jupiter is the largest planet in the solar system, and its great size and strong magnetic field make its boundary with the solar wind especially powerful and interesting to study.

Break it down

LEAD: Juno found Jupiter blocks the solar wind with a more complex system of plasma waves and shock regions than Earth uses.

KEY EVENT: Researchers discovered Jupiter relies on multi-frequency plasma waves and extra shocklets.

+- WHO: Scientists at the University of Iowa, using measurements from NASA's Juno spacecraft

+- WHAT: Jupiter's bow shock uses harmonics and shocklets to slow and heat solar particles

+- WHO SAID: Bill Kurth, an Iowa research scientist on the Juno mission since its 2011 launch

EVIDENCE: Earth's bow shock mainly uses waves around a single frequency and generally relies on its main bow shock alone.

WIDER IMPLICATION: The results could help scientists understand extreme shock waves from exploding stars.

Why it matters

Learning how Jupiter tames a much stronger solar wind helps scientists protect spacecraft and understand the violent shock waves around exploding stars.

Test what you remember

1. What spacecraft made the measurements used in this research?

- (A) Juno
- (B) Hubble
- (C) Cassini
- (D) Voyager

2. What is the solar wind?

- (A) A storm of dust around Mars
- (B) A stream of electrically charged particles continuously released by the Sun
- (C) Light reflected off the Moon
- (D) Heat rising from Earth's core

3. What does Earth's bow shock do to incoming solar particles?

- (A) It speeds them up toward the ground
- (B) It turns them into visible light
- (C) It traps them in Earth's oceans
- (D) It slows and heats them and redirects much of the solar wind around the planet

4. How does Jupiter's wave pattern differ from Earth's?

- (A) Earth uses shocklets and Jupiter does not
- (B) Both planets use exactly the same wave pattern
- (C) Jupiter has no plasma waves at all
- (D) Jupiter uses plasma waves with multiple frequencies, while Earth's waves are concentrated around a single frequency

5. What are shocklets?

- (A) Craters on Jupiter's surface
- (B) Bursts of light from the Sun
- (C) Smaller shock structures that slow the solar wind before it reaches the main bow shock
- (D) Tiny moons orbiting Jupiter

6. Why does Earth generally not need shocklets, according to Bill Kurth?

- (A) Because Earth's bow shock is larger than Jupiter's
- (B) Because Earth has no magnetic field
- (C) Because the solar wind never reaches Earth
- (D) Because the impact on Earth is not as powerful

Answer key (try the quiz first!): 1: A 2: B 3: D 4: D 5: C 6: D

Questions to think about

1. Bill Kurth's explanation: He said Earth does not need these extra steps because the impact there is not as powerful.

MY THOUGHTS (at least 20 words):